

# China Testing & Certification International Group Co., Ltd.

## Test Report

Report No.: CY2022B03C00101E

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| No.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Test Item                        | Standard Requirement<br>(Class A adhesive in Grade I)                                                                                                     | Test Result                                                                                                                                            | Item Conclusion | Test Method                            |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------------------------------------|
| 18                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Alkaline resistant medium effect | Compared with the control group, the strength does not decrease, and the concrete is damaged, and the appearance must not be cracked, peeled or blistered | Compared with the control group, the strength has not decreased, and the concrete is damaged, and the appearance has no cracks, peeling and blistering | Pass            | GB 50728-2011 Table 4.2.2-5 Appendix G |
| 19                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Acid resistant medium effect     | The concrete is damaged, and the appearance must not be cracked or degummed                                                                               | The concrete is damaged, without cracks and degumming in appearance                                                                                    | Pass            | GB 50728-2011 Table 4.2.2-5 Appendix G |
| (Blank)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                  |                                                                                                                                                           |                                                                                                                                                        |                 |                                        |
| <b>Remark:</b><br>1. This report is a safety appraisal report. According to the provisions of 3.0.8 in GB 50728-011, the qualification retention period of engineering reinforcement materials or products qualified for application safety appraisal is 4 years. According to the sampling rules stipulated in 3.0.3 of GB 50728-2011, 3 batches shall be sampled.<br>2. Test location: Guanzhuang.<br>3. The strength grade of the concrete substrate used in the test of No.18~19: C45. |                                  |                                                                                                                                                           |                                                                                                                                                        |                 |                                        |

End of the report

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## TECHNICAL EXPLANATION OF TESTING & ACCREDITATION AUTHORITIES

### Injectable Anchoring Epoxy Resin – HM-500

This document clarifies the technical standing, authority level, and professional relevance of the organizations involved in testing the HM-500 Injectable Anchoring Epoxy Resin.

HM-500 يوضح هذا المستند المكانة الفنية ومستوى الاعتماد والأهمية المهنية للجهات المشاركة في اختبار نظام التثبيت الإيبوكسي وذلك لدعم التقييم الفني واعتماد الجهات الاستشارية، 500.

### China Testing & Certification International Group (CTC)



China Testing & Certification International Group (CTC) is a state-owned testing, inspection, and certification organization operating under governmental supervision.

It is recognized as one of the highest-authority bodies in China for testing construction materials, engineering products, and infrastructure systems.

Testing performed by CTC confirms that product evaluation is conducted by a government-backed authority, independent from manufacturers or commercial laboratories.

تعد China Testing & Certification International Group (CTC) جهة مملوكة للدولة وتعمل تحت إشراف حكومي مباشر.

وتعتبر من أعلى الجهات سلطة في الصين لاختبار مواد البناء والمنتجات الهندسية وأنظمة البنية التحتية.

يؤكد أن التقييم تم من خلال جهة CTC إجراء الاختبارات بواسطة حكومية معتمدة ومستقلة عن المصنعين أو المختبرات التجارية.

### National Research Center of Testing Techniques for Building Materials



This institution operates as a national research-level authority specializing in building materials, structural systems, and construction safety evaluation.

Its role extends beyond routine compliance testing to scientific validation, performance assessment,

يعمل هذا المركز كجهة بحثية وطنية متخصصة في مواد البناء والأنظمة الإنشائية وتقييم السلامة الهيكلية.

ويمتد دوره إلى التحقق العلمي وتقييم الأداء وتحليل السلوك الإنشائي وحدود الفشل.

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and failure behavior analysis.

Involvement of a national research institution confirms that testing is supported by engineering research rather than commercial testing only.

مشاركة جهة بحثية وطنية تؤكد أن الاختبارات مدعومة ببحث علمي وتحليل هندسي وليس اختبارًا تجاريًا فقط.

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## China National Accreditation Service (CNAS)



CNAS is the official national accreditation body of China, responsible for accrediting testing and calibration laboratories in accordance with ISO/IEC 17025.

Accreditation verifies laboratory competence, calibrated equipment, qualified personnel, validated test methods, and audited quality systems.

CNAS operates at the same professional level as leading global accreditation bodies such as UKAS and DAKKS.

الجهة الوطنية الرسمية لاعتماد مختبرات الاختبار CNAS تُعد ISO/IEC 17025 والمعايرة في الصين وفقًا لمعيار

يضمن هذا الاعتماد كفاءة المختبر ودقة الأجهزة وتأهيل الكوادر الفنية وصحة طرق الاختبار ونظم الجودة

UKAS مماثلًا لجهات اعتماد عالمية مثل CNAS ويُعد اعتماد DAKKS.

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## International Laboratory Accreditation Cooperation – ILAC-MRA



ILAC-MRA represents a global mutual recognition arrangement between national accreditation bodies.

It ensures international acceptance of test results issued by accredited laboratories.

This framework reduces the need for duplicate testing in multinational engineering and construction projects.

اتفاقية اعتراف متبادل دولية بين هيئات الاعتماد ILAC-MRA تمثل الوطنية

وتضمن القبول الدولي لنتائج الاختبارات الصادرة عن مختبرات معتمدة

تتيح هذه المنظومة الاعتماد على النتائج دون الحاجة إلى إعادة الاختبار في المشاريع متعددة الجنسيات

## China Metrology Accreditation (CMA)



CMA is a mandatory governmental accreditation required for laboratories issuing legally valid test reports in China.

اعتمادًا حكوميًا إلزاميًا للمختبرات التي تصدر تقارير CMA تُعد اختبار ذات صفة قانونية داخل الصين.

The CMA mark confirms legal validity, regulatory recognition, and enforceable accountability of test reports.

إلى أن التقرير يتمتع بقيمة قانونية وتنظيمية CMA وتشير علامة ملزمة إلى جانب قيمته الفنية.

## Overall Technical Assessment

The HM-500 epoxy anchoring system was tested under a framework combining state-owned testing authorities, national research institutions, and internationally aligned accreditation systems.

تم إجراء اختبارات نظام التثبيت الإيبوكسي معتمده، يجمع بين جهات اختبار حكومية، ومؤسسات بحثية وطنية، وأنظمة اعتماد دولية متوافقة مع المعايير العالمية.

This confirms that the reported results are independent, scientifically validated, legally recognized, and internationally accepted for professional engineering evaluation and approval.

ويؤكد ذلك أن نتائج الاختبارات مستقلة، مدعومة علميًا، معترف بها قانونيًا، ومقبولة دوليًا، وقابلة للاعتماد في التقييمات الهندسية الاحترافية.



# Submission Review and Comment Sheet



## REVIEW NOTICE

| Green Line          |                                                                    |                   |                         |
|---------------------|--------------------------------------------------------------------|-------------------|-------------------------|
| RCS Number          |                                                                    |                   |                         |
| Transmittal No      | NBS-PTJ-ECG-MAT-CVL-059                                            | Document Number   | NBS-PTJ-ECG-MAT-CVL-059 |
| Transmittal Date    | 26/11/2025                                                         | Receiving Date    | 01/12/2025              |
| Contractor Name     | PETROJET                                                           | Document Revision | 00                      |
| Designer            | ECG                                                                | Location          | NOUBARYA STATION        |
| Transmittal Subject | MAR – INJECTABLE EPOXY - HM-500 (STRUCTURAL BAR IMPLANTING - HORSE |                   |                         |
| Document Title      | MATERIAL APPROVAL REQUEST                                          |                   |                         |

### The following Conditional Notes apply to this Review:

- Where an RCS replies to an advance submission, any approval noted is not effective until the formal submission has been received by ER and any progress of the Works shall be at the Contractor's own risk
- Approval or comments by the ER of any equivalent or superior designs, products, services, standards or any other deliverables to those designated in the Contract which are proposed by the Contractor, shall not constitute a variation nor entitle the Contractor to any additional payment or extension of time.
- The status given by the Employer's Representative /General Consultant for this submission does not exonerate or relieve the Contractor from any obligation or liability hereunder. Each such status is given in reliance upon, and subject to the performance by the Contractor of its Works in accordance with and pursuant to the terms, covenants, provisions and conditions of its Contract. Without limitation to the generality of the foregoing, neither the review nor the Approved status of any submittal or portion thereof by the Employer's Representative / General Consultant shall relieve the Contractor of any of its responsibilities whatsoever for the accuracy or integrity of the submittal, or the ability of the Contractor to meet any conditions specified in the submittal. While Employer's Representative / General Consultant retains rights of Approval for all submittals, the areas in which it retains Approved rights is limited to the Employer's Representative / General Consultant scope of work, for the information provided by the Contractor.
- The Contractor may include many documents in one submission (Transmittal form) however where possible, it is ER policy to reply individually to each listed submitted document.
- If appropriate, ER may respond to many documents received on one submission (Transmittal Form) where one response will be applied to the entire submission package.
- Document number shall be a document in the transmission list (usually the first number listed)
- This Review comments comprises all of the pages listed in the footer below
- The Review Notice Status shown below applies to all documents described in the Comments Sheet
- Major (M) classified comments are issued when the referenced document submission fails to meet or describe a requirement in respect of how the submission requirement is organised, implemented or maintained and without correction will introduce a major risk and will jeopardise the effectiveness of the submission if implemented
- Minor (m) classified comments are issued when the referenced document submission fails to meet or describe a requirement in respect of how the submission requirement is organised, implemented or maintained and without correction may jeopardise the effectiveness of the submission.
- The Contractor must state if their response to ER Comments have been incorporated (I) into a revised document or, shall not be incorporated (NI) as described
- Any ASCII character may be used to denote data entry into the fields for M, m, I or NI
- Unresolved comments shall be incremented by a decimal index and all related comments shall remain open until the subject is resolved
- Project procedure EET-SYS-MPL-OPMDOC-GL-CON-1001 provides detailed definition of "Summary Meaning" below

| STATUS OF THE REVIEW NOTICE |                                                                                 | APPROVED WITH COMMENTS |
|-----------------------------|---------------------------------------------------------------------------------|------------------------|
| Status                      | Summary Meaning                                                                 |                        |
| APPROVED                    | No objection to proceed                                                         |                        |
| APPROVED WITH COMMENTS      | No objection if the Contractor incorporates comments and resubmits for approval |                        |
| REJECTED                    | Revise and resubmit after fulfilling the comments                               |                        |
| RECEIPT FOR INFORMATION     | ER receipt only no action required                                              |                        |